

Azure Cloud Fundamentals and Data Pipeline Implementation using Azure Data Factory

Name: Manas Arora

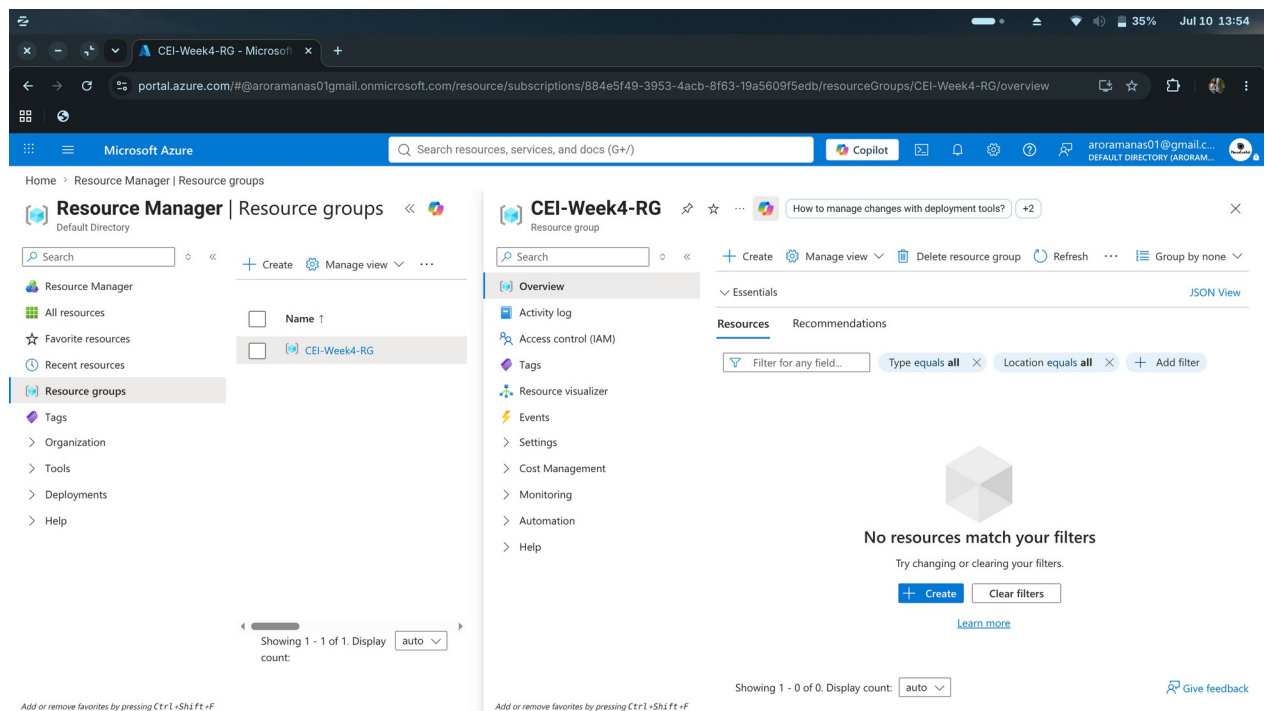
Intern ID: CT_CSI_DE_1089

Objective

Understand Azure cloud fundamentals and build an end-to-end data pipeline using Azure Storage Account and Azure Data Factory.

Step 1 – Resource Group

Created an Azure Resource Group to organize all resources used in the assignment.



Step 2 – Storage Account

Provisioned an Azure Storage Account to securely store the dataset and outputs.

Home > Storage center | Blob Storage

Create a storage account

Subscription *

Azure for Students

Resource group *

CEI-Week4-RG

[Create new](#)

Instance details

Storage account name * ⓘ

manasceiweek4

Region * ⓘ

(Asia Pacific) Central India

[Deploy to an Azure Extended Zone](#)

Preferred storage type

Azure Blob Storage or Azure Data Lake Storage

This helps us provide relevant guidance. It doesn't restrict your storage to this resource type. [Learn more](#)

Performance * ⓘ

☒

Standard: Recommended for most scenarios (general-purpose v2 account)

☐

Premium: Recommended for scenarios that require low latency.

Redundancy * ⓘ

Locally redundant storage (LRS)

Previous

Next

Review + create

Step 3 – Blob Containers

Set up Blob containers to act as source and destination repositories for the data pipeline.

Home > manasceiweek4

manasceiweek4 | Containers

Storage account

Search

+ Add container Upload Refresh Delete Change access level Restore containers Edit columns

Showing all 3 items

☐	Name	Last modified	Anonymous access level	Lease state	Storage connector
☐	logs	10/07/2026, 14:28:26	Private	Available	-
☐	destination	10/07/2026, 14:31:58	Private	Available	-
☐	source	10/07/2026, 14:30:12	Private	Available	-

Successfully created storage container
Successfully created storage container 'destination'.

Overview
Activity log
Tags
Diagnose and solve problems
Access Control (IAM)
Data migration
Events
Storage browser
Storage Mover
Partner solutions
Resource visualizer
Data storage
Containers
Classic file shares
Queues
Tables

Add or remove favorites by pressing Ctrl+Shift+F

Step 4 – CSV Uploaded

Uploaded the raw CSV dataset into the source Blob container.

Home > manasceiweek4 | Containers

source

Container

Search

+ Add Directory Upload Change access level Refresh Delete Copy Paste Rename Acquire lease Break lease Edit columns

source

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	superstore.csv	10/07/2026, 14:33:33	Hot (Inferred)	Block blob	2.18 MiB	Available

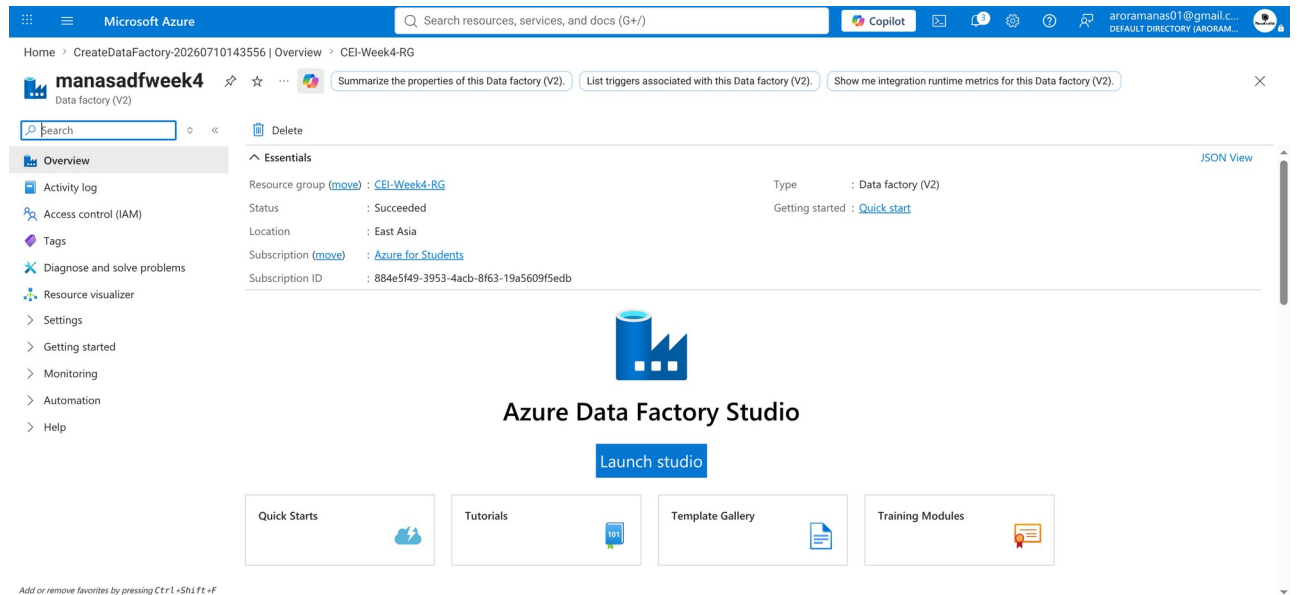
Successfully uploaded blob(s)
Successfully uploaded 1 blob(s).

Overview
Diagnose and solve problems
Access Control (IAM)
Settings

Add or remove favorites by pressing Ctrl+Shift+F

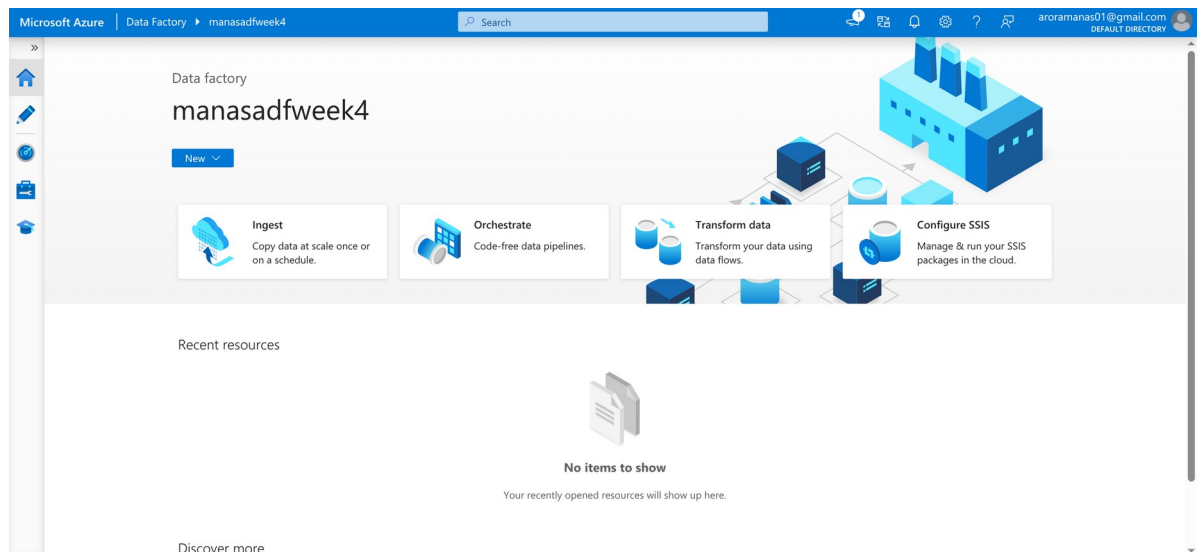
Step 5 – ADF Overview

Provisioned an Azure Data Factory instance to orchestrate the data movement.



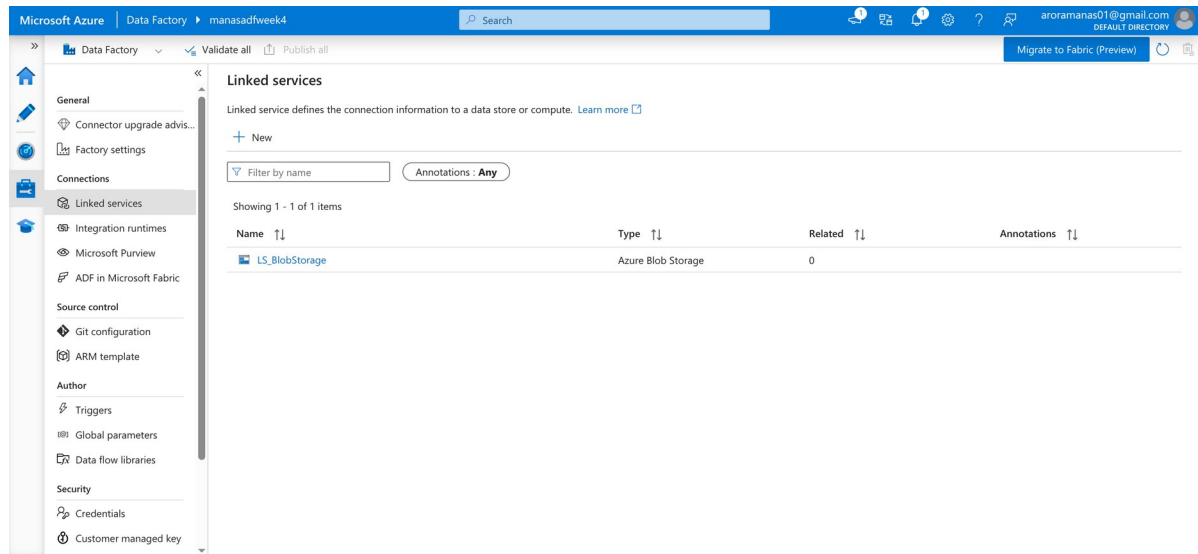
Step 6 – ADF Studio Home

Launched the Azure Data Factory Studio to design and manage the data pipeline.



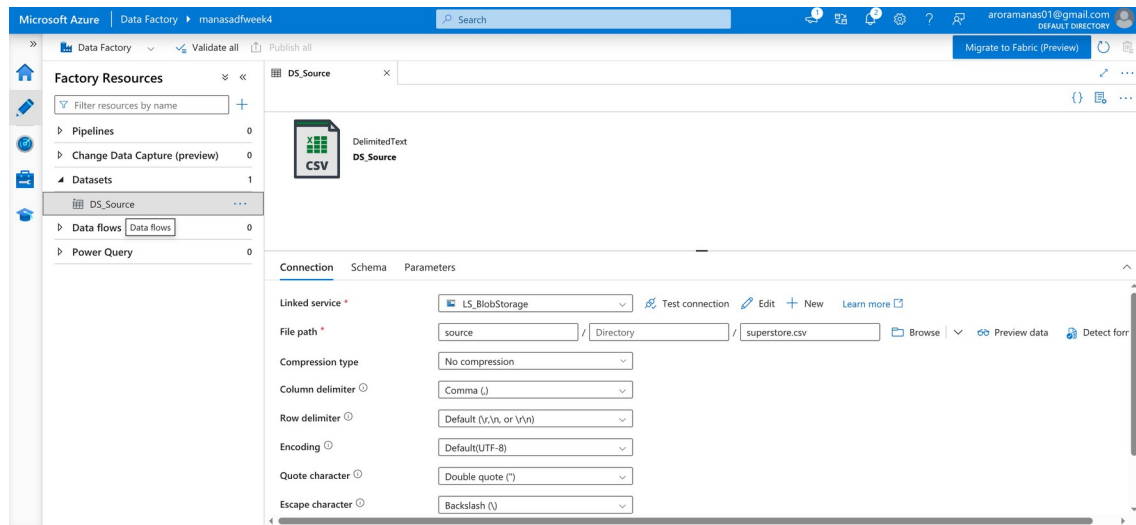
Step 7 – Linked Service

Created a Linked Service to securely connect Azure Data Factory with the Storage Account.



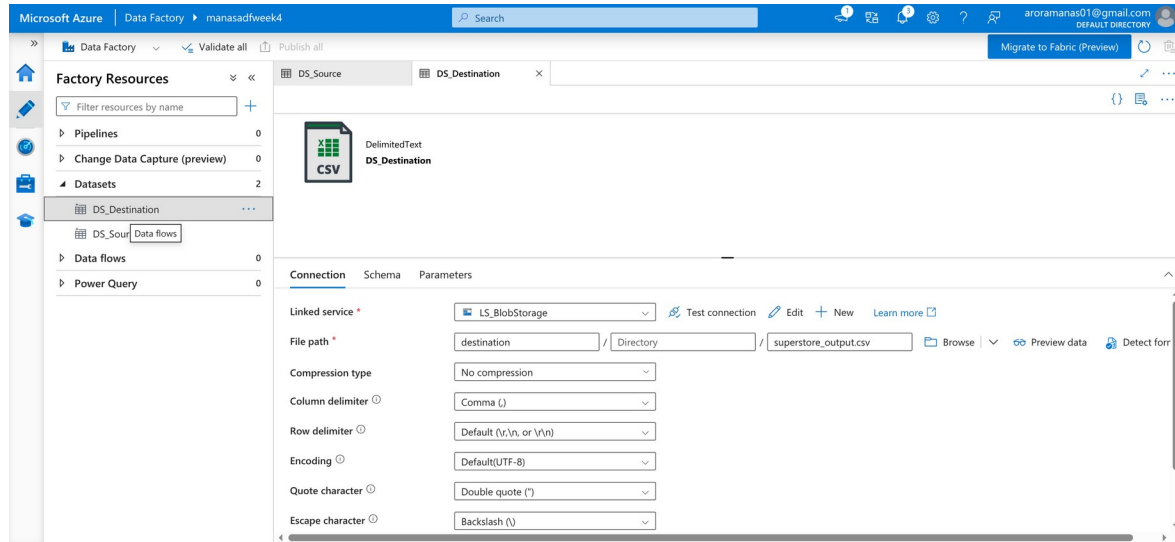
Step 8 – Source Dataset

Configured a source dataset in ADF pointing to the uploaded CSV file in Blob storage.



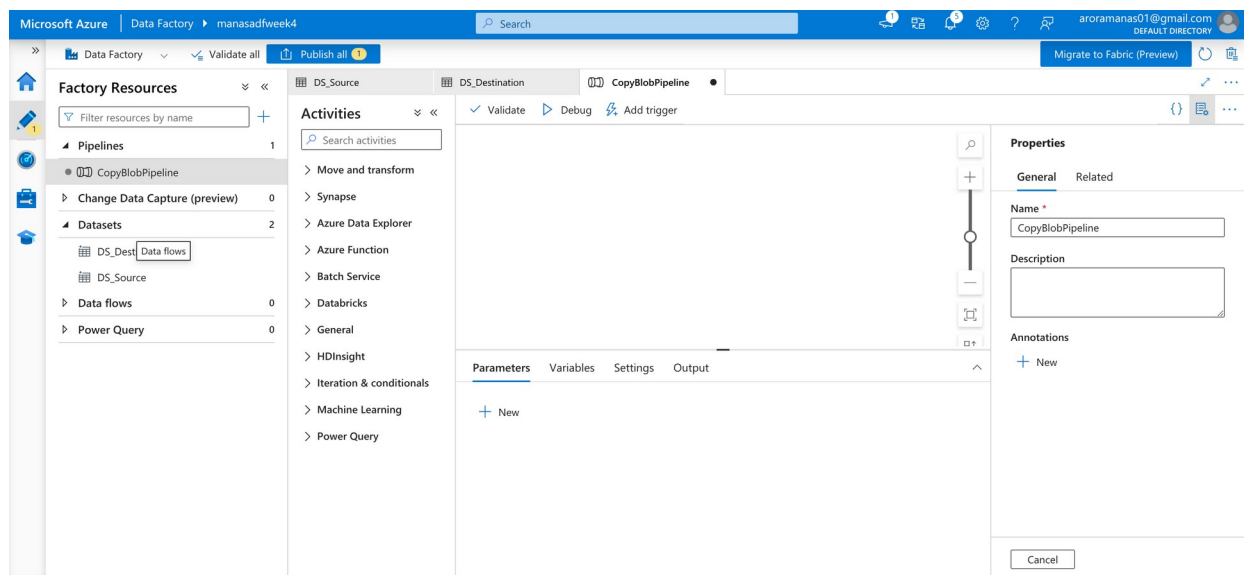
Step 9 – Destination Dataset

Configured a destination dataset to define where the processed data will be saved.



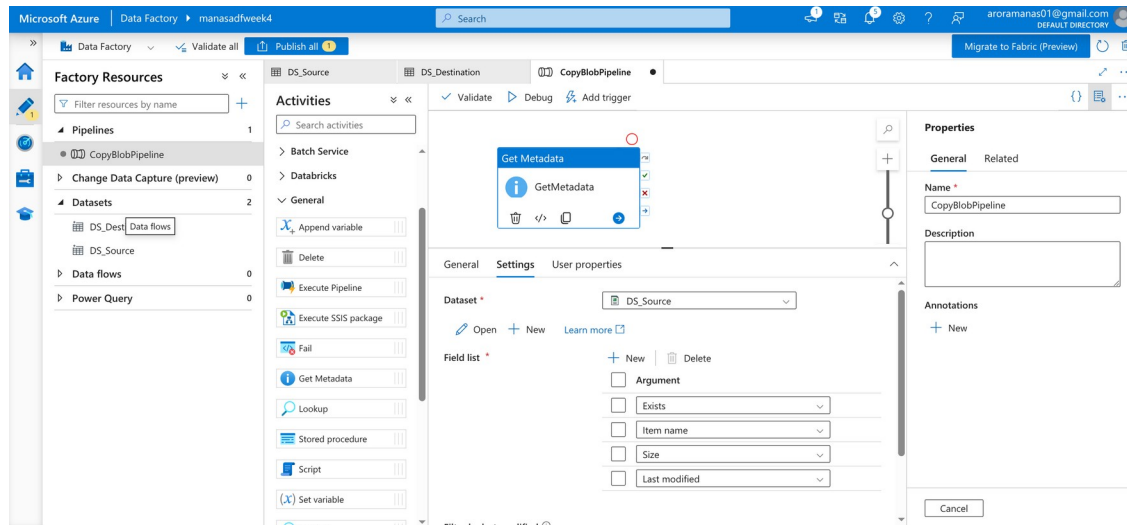
Step 10 – Pipeline Overview

Created a new data pipeline to connect the source and destination datasets.



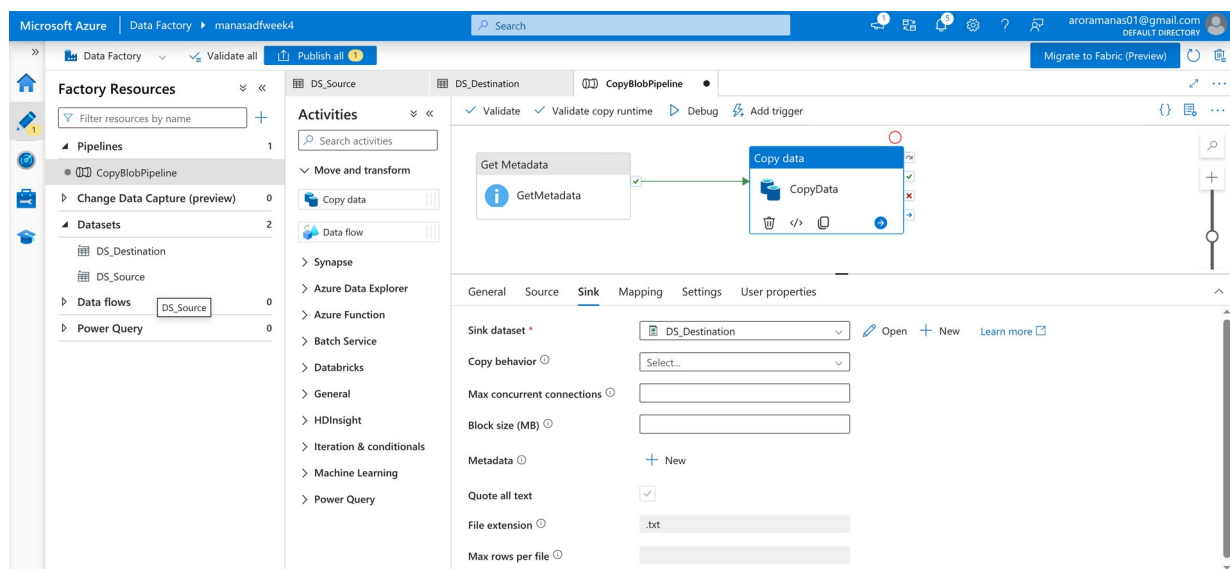
Step 11 - Get Metadata

Implemented a Get Metadata activity to validate the structure and existence of the source file.



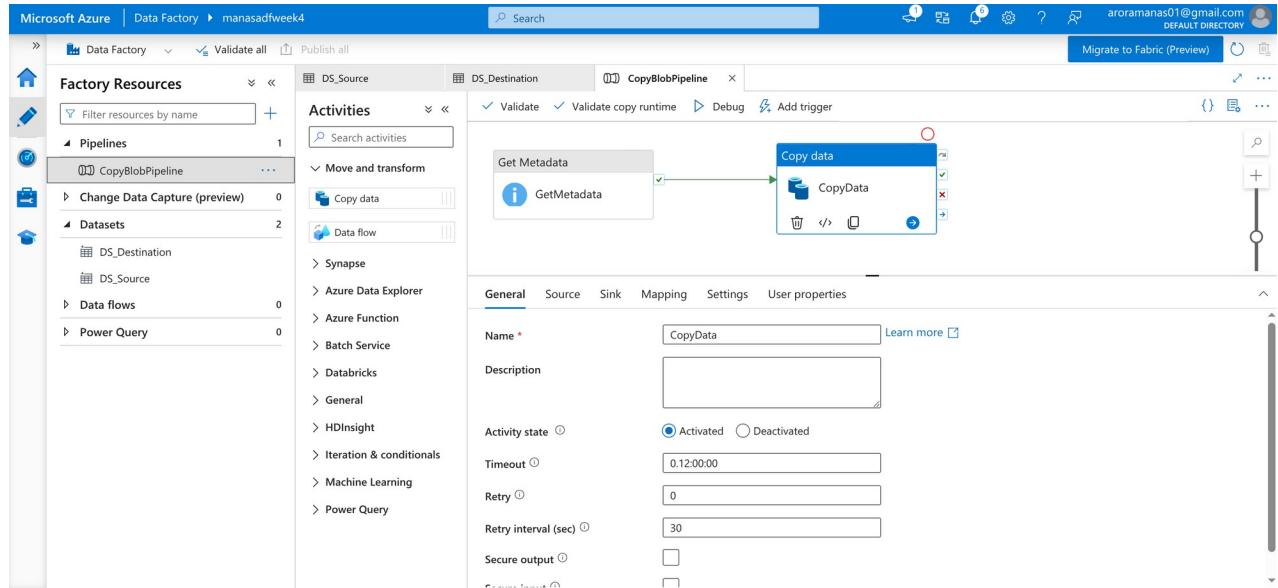
Step 12 - Copy Data Pipeline

Added a Copy Data activity to securely transfer the data from the source to the destination container.



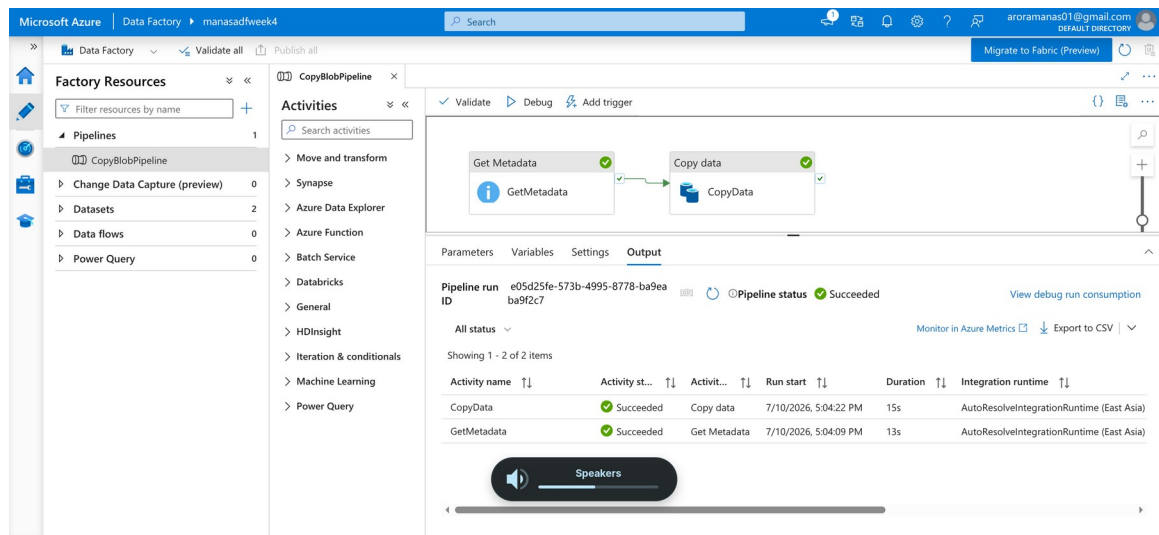
Step 13 – Publish Success

Published all pipeline components to save and deploy the configurations in ADF.



Step 14 – Debug Success

Executed a debug run to validate the pipeline flow without errors.



Step 15 – Pipeline Run

Triggered a manual pipeline run to process the data in the production environment.

The screenshot displays the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane shows the 'CopyBlobPipeline' under the 'Pipelines' section. The main area shows the pipeline's execution details. A notification at the top right states 'Run Succeeded' with the message 'Successfully ran CopyBlobPipeline (Pipeline)'. Below this, the 'Output' tab shows the pipeline run ID 'e05d25fe-573b-4995-8778-ba9ea-ba9f2c7' and its status 'Succeeded'. A table lists the activities and their results:

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
CopyData	✓ Succeeded	Copy data	7/10/2026, 5:04:22 PM	15s	AutoResolveIntegrationRuntime (East Asia)
GetMetadata	✓ Succeeded	Get Metadata	7/10/2026, 5:04:09 PM	13s	AutoResolveIntegrationRuntime (East Asia)

Step 16 – IAM Roles

Configured Identity and Access Management (IAM) roles to ensure secure and least-privilege access.

The screenshot shows the Microsoft Azure Storage Center interface. The left pane displays 'All storage resources' with 'manasceiweek4' selected. The main area shows the 'Access Control (IAM)' settings for the selected storage account. The 'Role assignments' tab is active, showing a table of role assignments for the subscription.

Name	Type	Role	Scope	Cc
Owner (2)				
MANAS ARORA a941248c-4516...	User	Owner	Subscription (inhe...	Ni
MANAS ARORA a941248c-4516...	User	Owner	Subscription (inhe...	Ni
Reader (1)				
manasceiweek4 4daff759-3f29...	Managed identity	Reader	This resource	Ni

Step 17 - Monitor

Monitored the pipeline execution using the ADF monitoring dashboard to track activity success.

The screenshot shows the Microsoft Azure Data Factory monitoring dashboard for the 'CopyBlobPipeline - Activity runs'. The left sidebar contains navigation options: Dashboards, Runs, Pipeline runs, Trigger runs, Change Data Capture (previ...), Runtimes & sessions, Integration runtimes, Data flow debug, Notifications, and Alerts & metrics. The main area displays a Gantt chart of the pipeline run, showing two activities: 'Get Metadata' and 'Copy data', both marked as successful with green checkmarks. Below the Gantt chart, the 'Activity runs' section shows a table of activity runs for the pipeline run ID 'd255c4e1-68d9-43c4-8dbf-c5fd8843a93e'. The table lists two activities: 'CopyData' and 'GetMetadata', both with a status of 'Succeeded'. The 'CopyData' activity ran for 16s, and the 'GetMetadata' activity ran for 12s. Both activities used the 'AutoResolveIntegrationRuntime (East Asia)'.

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime	User prop...	Activity run ID
CopyData	Succeeded	Copy data	7/10/2026, 5:06:19 PM	16s	AutoResolveIntegrationRuntime (East Asia)		709d29ee-da40-4
GetMetadata	Succeeded	Get Metadata	7/10/2026, 5:06:07 PM	12s	AutoResolveIntegrationRuntime (East Asia)		c08bd0f4-a8bc-4

Step 18 - Output File

Verified the successful transfer of the dataset in the destination Blob container.

The screenshot shows the Microsoft Azure Storage Explorer interface for the 'destination' container. The left sidebar contains navigation options: Overview, Diagnose and solve problems, Access Control (IAM), and Settings. The main area displays the 'destination' container, which is currently empty. The 'Authentication method' is set to 'Access key (Switch to Microsoft Entra user account)'. The 'Add filter' section shows a search filter 'Search blobs by prefix (case-sensitive)' and a dropdown menu 'Only show active blobs'. The table below shows the contents of the container, which are currently empty.

Name	Last modified	Access tier	Blob type	Size	Lease state
superstore_output.csv	10/07/2026, 17:06:32	Hot (Inferred)	Block blob	2.18 MiB	Available

Folder Structure

Week-4/

```
|
|— README.md
|— Week4_Assignment.ipynb
|— DataSet/
|   |— superstore.csv
|— Resource/
|   |— Week 4 Task.pdf
|— Screenshots/
|   |— (18 Screenshots)
```

File Description

- [Week4_Assignment.ipynb](#): The Jupyter Notebook version of this README, documenting all steps.
- [DataSet/superstore.csv](#): The raw dataset uploaded to Azure Blob Storage.
- [Resource/Week 4 Task.pdf](#): The official assignment tasks and guidelines for Week 4.
- **Screenshots/**: Directory containing all step-by-step pipeline execution screenshots.

Technologies Used

- Microsoft Azure
- Azure Storage Account
- Azure Blob Storage
- Azure Data Factory (ADF)
- Azure IAM
- CSV Dataset

Azure Services Used

- Resource Group
- Storage Account
- Blob Container
- Azure Data Factory
- Linked Service
- Dataset
- Pipeline
- Get Metadata
- Copy Data
- IAM

Mini Project

An end-to-end Azure Data Factory pipeline was implemented which: - Reads a CSV file from Azure Blob Storage - Validates metadata using Get Metadata - Copies the file using Copy Data - Stores the processed file into the destination Blob container - Successfully executes the pipeline

Learning Outcomes

- Set up Azure Resource Groups and Storage Accounts.
- Configured Blob containers for raw and processed data.
- Created Linked Services and Datasets in Azure Data Factory (ADF).
- Used Get Metadata to validate files before moving them.
- Built and ran a data pipeline using the Copy Data activity.
- Assigned IAM roles for access control.

Conclusion

This assignment gave me hands-on experience with Azure Data Factory and Blob Storage. I successfully built an end-to-end pipeline to validate and transfer files between storage containers. The pipeline ran perfectly, and managing IAM roles helped me understand basic security in Azure. Everything worked exactly as expected without any issues.